

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects.
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 10

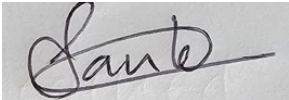
Total Marks: 30

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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1. Unified retail customer architecture

Fragmented online and offline records should be combined into a common customer view so real-time events and history can support personalization.

- Integrate POS, online, CRM and interaction data.
- Use streaming for live events.
- Use batch processing for historical analytics.
- Apply common customer IDs and quality rules.
- Use recommendation models with secure access.

Industry Big Data Architecture

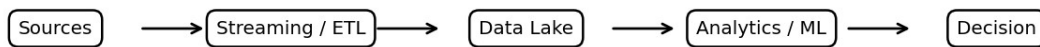


2. Distributed computing for scientific research

Large scientific datasets benefit from parallel execution because independent portions can be processed simultaneously across nodes.

- Divide datasets into parallel tasks.
- Execute tasks across multiple compute nodes.
- Process data close to distributed storage.
- Scale horizontally by adding nodes.
- Recover failed tasks without restarting all work.

Industry Big Data Architecture



3. Digital payment fraud detection

Fraud prevention must react quickly while considering historical behavior. A streaming pipeline with predictive scoring fits the requirement.

- Stream payment events for immediate analysis.
- Combine live features with historical patterns.
- Use rules plus ML/anomaly detection.
- Trigger alerts when risk crosses a threshold.
- Maintain low latency, security, audit and model monitoring.

Industry Big Data Architecture



4. Multimedia data management

Video, audio and images require significant storage and I/O and need metadata for efficient retrieval. Distributed storage improves scalability.

- Use scalable distributed/object storage.
- Store metadata for search and classification.
- Apply compression and lifecycle policies.
- Use distributed processing for transformation.
- Protect multimedia data with access controls and encryption.

Industry Big Data Architecture

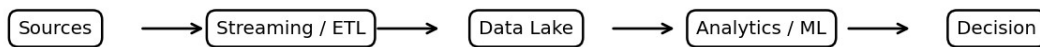


5. Cloud big data disaster recovery

Resilience requires redundant services, independent backups and tested recovery. These reduce downtime and data loss.

- Replicate critical data across failure domains.
- Use redundant services and automated failover.
- Maintain versioned backups.
- Define recovery objectives and monitor health.
- Test restoration regularly.

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6. Data quality issues

Inconsistency, duplication and missing values can distort analytics and ML results. Governance should detect and correct problems early.

- Standardize formats and definitions.
- Detect and remove duplicates.
- Validate missing or invalid values.
- Track lineage and data ownership.
- Monitor quality continuously.

Industry Big Data Architecture



7. Streaming recommendation system

Viewing behavior provides key signals for personalization. Distributed processing is needed to generate features and recommendations at scale.

- Collect views, searches, ratings, skips and sessions.
- Generate features with distributed processing.
- Use collaborative, content-based or hybrid models.
- Rank recommendations using context features.
- Use engagement feedback for retraining.

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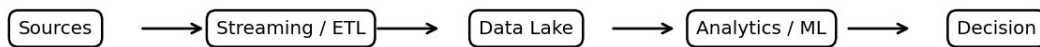


8. Cloud computing in big data

Cloud platforms provide elastic resources and managed services, but cost, security and dependency must be controlled.

- Scale compute and storage with demand.
- Use pay-as-you-use resources with monitoring.
- Adopt managed analytics, storage, databases and ML.
- Apply strong identity and security controls.
- Manage vendor dependence, transfer cost, latency and governance.

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9. Transportation and logistics analytics

Real-time location and traffic data can improve routing and operational decisions. Streaming analytics keeps decisions current.

- Collect GPS, traffic, vehicle and delivery events.
- Process live data for route/congestion analysis.
- Use geospatial and travel-time prediction.
- Optimize fuel use, idle time and routes.
- Provide dashboards and operational alerts.

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10. Ethical and legal big data challenges

Responsible analytics requires privacy, transparency, fairness and compliance across the data lifecycle.

- Collect only necessary data for legitimate purposes.
- Provide consent and transparency.
- Use encryption, access control and audit logs.
- Evaluate models for bias and unfair profiling.
- Maintain compliance, user rights, retention and accountability.

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